

ISTRES AP, France

WMO: 076470

Lat: 43.523N

Lon: 4.922E

Elev: 82

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.8	31.4	14.7	11.6	35.3	19.2	14.5	37.3	43.5	44.3	39.0	44.0	11.7	350	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	91.8	69.3	89.4	68.4	87.1	68.0	74.2	83.5	72.7	82.1	71.3	80.8	12.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.6	116.9	78.2	69.8	109.9	77.2	68.1	103.7	76.2	37.7	84.2	36.3	82.0	35.1	80.8	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
36.8	31.9	27.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			25.5	96.9	2.9	2.8	23.4	98.9	21.7	100.6	20.1	102.1	18.0	104.1
			23.5	76.3	3.4	1.5	21.0	77.4	19.0	78.3	17.2	79.1	14.7	80.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	45.1	46.1	52.0	57.2	64.2	72.4	76.7	75.9	69.0	61.9	51.9	46.6
	DBStd	12.40	5.61	5.94	5.12	4.69	4.86	5.25	4.11	3.98	4.69	5.78	6.29	5.70
	HDD50	529	170	129	37	4	0	0	0	0	0	3	51	135
	HDD65	2968	617	530	402	236	72	6	0	0	16	126	392	571
	CDD50	4182	18	19	101	220	441	672	828	802	570	373	109	29
	CDD65	1145	0	0	0	2	47	228	363	337	137	31	0	0
	CDH74	9697	0	0	2	29	355	1985	3416	2949	865	96	0	0
	CDH80	3370	0	0	0	1	61	676	1359	1089	178	6	0	0
(14) Wind	WSAvg	11.8	12.3	13.0	12.6	12.3	11.7	11.5	12.0	10.9	10.9	11.0	12.0	11.8
(15) Precipitation	PrecAvg	23.9	2.1	1.4	1.3	2.2	1.8	1.2	0.7	1.2	3.6	3.5	3.0	1.9
	PrecMax	34.2	6.5	4.7	3.7	5.6	4.2	3.9	2.3	4.3	9.4	6.2	8.7	6.6
	PrecMin	14.6	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.2	0.6	1.5	0.7	0.0
	PrecStd	5.5	1.6	1.2	1.0	1.3	1.1	1.0	0.7	0.9	2.0	1.5	2.1	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	65.5	72.2	77.8	86.5	94.6	94.8	95.0	87.9	79.8	69.5	62.9
		MCWB	53.3	52.4	55.5	60.0	65.8	71.5	69.6	69.9	67.3	64.4	60.4	56.0
	2%	DB	59.1	61.0	67.9	73.6	81.0	90.5	91.8	91.3	84.4	76.7	66.6	60.5
		MCWB	52.2	50.9	54.1	58.5	63.1	68.6	69.2	69.5	66.6	65.0	59.7	54.5
	5%	DB	56.8	58.6	64.8	70.4	77.8	87.1	89.6	88.3	81.8	74.1	64.3	58.3
		MCWB	51.0	50.2	53.0	56.8	61.8	67.5	68.4	68.9	66.2	63.8	58.2	53.0
	10%	DB	54.7	56.1	62.1	67.6	75.0	83.8	87.3	86.0	79.1	71.6	62.1	56.1
		MCWB	49.4	49.1	52.1	55.4	61.0	66.5	68.0	68.2	65.2	62.8	56.6	51.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	54.8	58.3	63.1	69.0	74.3	76.2	76.7	72.6	69.2	63.5	58.1
		MCDB	60.1	60.0	65.9	71.4	78.2	87.6	82.8	85.1	78.9	74.3	67.0	61.0
	2%	WB	53.8	53.1	56.2	60.3	66.3	71.9	74.2	74.4	70.8	67.5	60.9	55.7
		MCDB	57.9	58.6	63.9	69.7	75.0	82.7	83.4	83.3	78.0	73.0	65.1	59.3
	5%	WB	51.9	51.6	54.8	58.5	64.4	70.2	72.3	73.0	69.3	65.9	59.2	53.9
		MCDB	56.1	57.1	62.1	67.2	74.0	81.1	82.4	81.9	76.7	71.7	63.5	57.4
	10%	WB	50.0	49.8	53.4	57.1	62.7	68.6	70.8	71.6	67.8	64.1	57.2	52.0
		MCDB	53.9	55.1	60.5	64.7	71.6	79.8	81.5	80.3	75.5	69.8	61.4	55.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	15.9	17.7	18.1	18.7	19.6	19.6	19.6	18.3	16.2	14.4	14.2
		MCDBR	16.7	20.2	22.1	22.1	23.0	23.4	22.2	22.8	21.3	18.2	16.1	16.0
		MCWBR	11.4	12.4	11.6	10.7	10.0	8.9	8.2	8.7	9.3	9.1	10.5	11.1
		MCDBR	14.9	18.3	19.1	20.0	21.0	21.2	19.3	19.5	18.4	15.4	14.3	13.8
	5% WB	MCWBR	11.2	12.5	11.1	10.9	10.5	9.2	8.7	9.6	9.9	9.1	10.3	10.7
(40) Clear-Sky Solar Irradiance	taub		0.328	0.346	0.383	0.400	0.410	0.410	0.403	0.401	0.390	0.390	0.353	0.324
	taud		2.468	2.414	2.322	2.298	2.320	2.348	2.369	2.378	2.397	2.391	2.443	2.483
	Ebn at Noon		254	268	270	274	274	274	271	265	248	240	244	244
	Edh at Noon		23	29	36	40	40	39	38	36	33	30	24	21
(44) All-Sky Solar Radiation	RadAvg		566	877	1300	1703	2054	2368	2360	2021	1536	970	625	491
	RadStd		59	84	95	115	136	95	86	64	84	84	69	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	+0.52	N/A	N/A	+1.17	N/A	N/A	N/A	N/A	N/A	+83

Nomenclature: See separate page